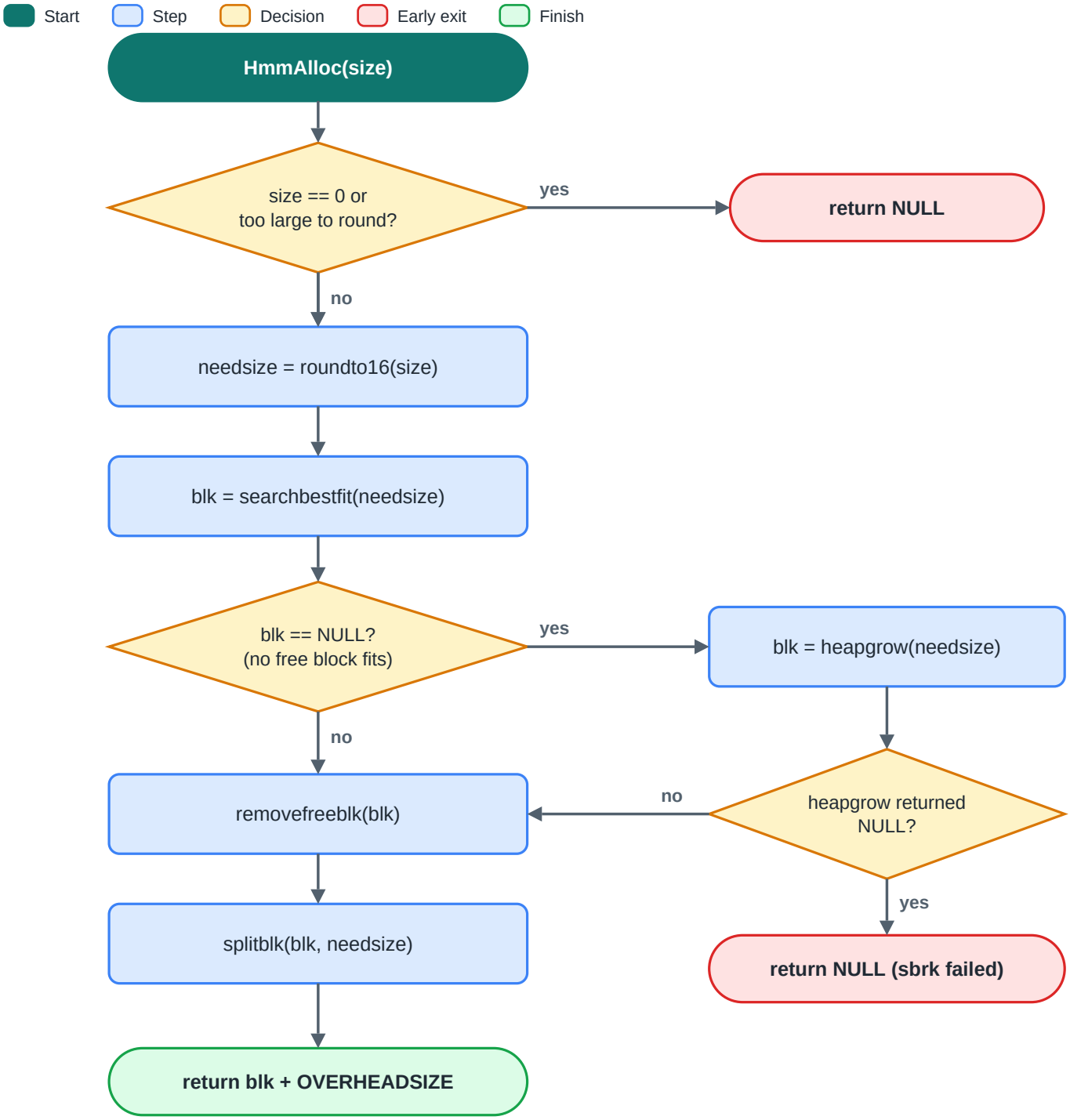
HmmAlloc(size)

Allocation flow: round, search, grow if needed, split, return

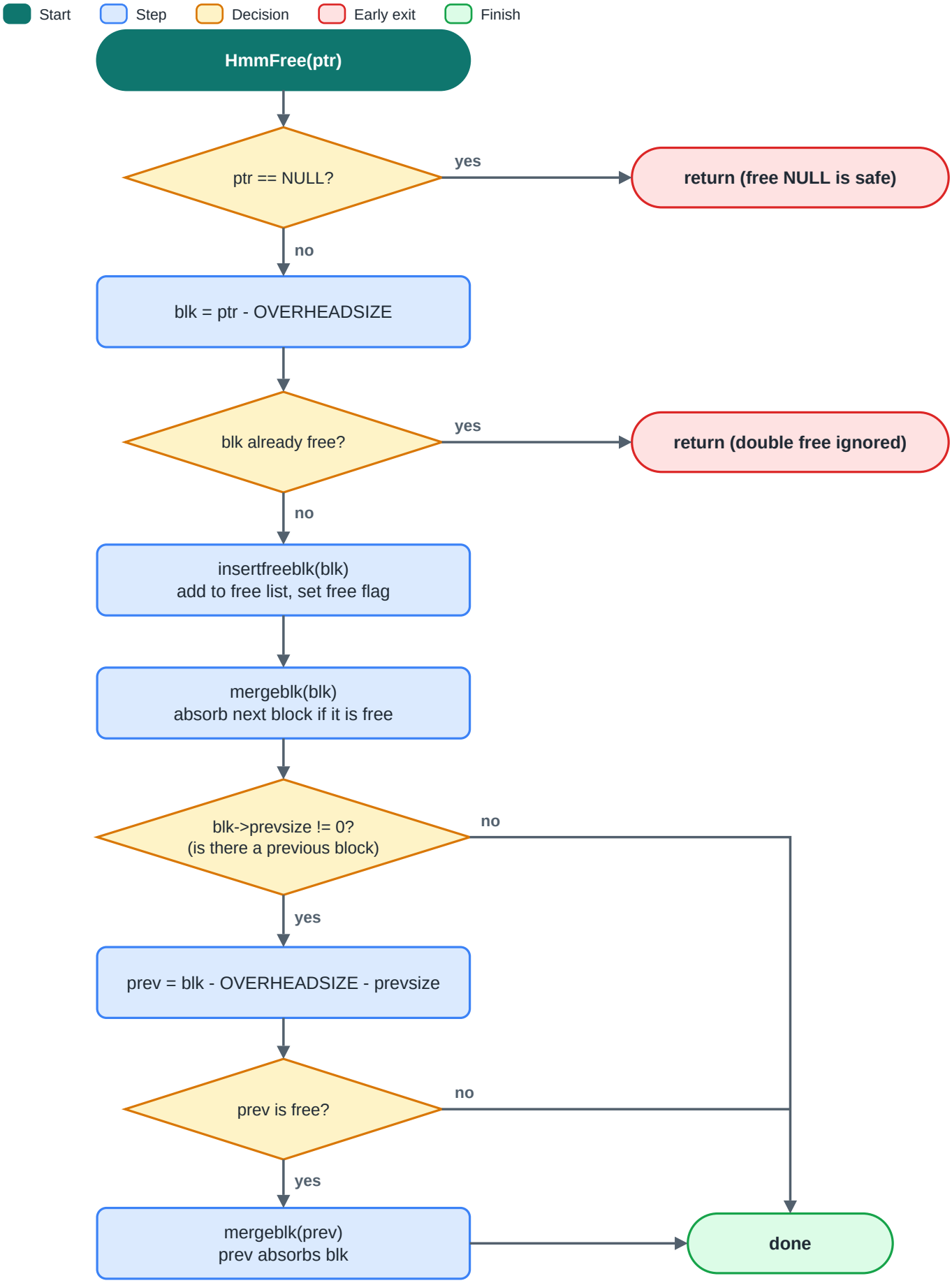


Why each step is there

- Rounding happens once here, so search, split and heapgrow all use the same size.
- Best fit scans the free list and keeps the smallest block that still fits.
- heapgrow asks sbrk for at least 128 KB and merges with a free last block.
- removefreeblk runs before splitblk, so the user's block is already out of the list.
- splitblk only cuts off a leftover that can hold a 16-byte header plus 16 data bytes.
- The returned pointer is block + 16, which keeps it 16-byte aligned.

HmmFree(ptr)

Free flow: validate, insert, then coalesce with next and previous block



Why this order works

- mergeblk(blk) runs first, while blk is still a valid header.
- mergeblk(prev) finds blk as its free next neighbour and absorbs it.